MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 06 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

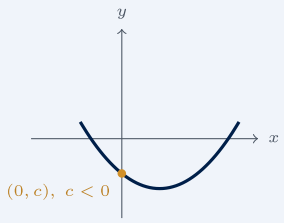
MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. If the graph of quadratic polynomial $ax^2 + bx + c$ cuts the negative direction of the y-axis, then the sign of c is: [1]

(A) positive (B) negative (C) zero (D) depends on a

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)



Graphical Reason: At $x = 0$, $y = c$. Because the intersection lies below the origin along the negative y-axis, $c < 0$ regardless of whether $a > 0$ or $a < 0$.

We know that the graph of $y = ax^2 + bx + c$ intersects the y-axis at $x = 0$:

Formula Used: y-intercept $= (0,\ c)$

Cutting the negative direction of the y-axis means $y < 0$ at $x = 0$, so $c < 0$, i.e., c is $\boxed{\text{negative}}$. [1M]

Q2. If the graph of $y = ax^2 + bx + c$ opens upwards, then: [1]

(A) $a > 0$ (B) $a < 0$ (C) $a = 0$ (D) $c > 0$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know the leading-coefficient orientation rule for a quadratic parabola $y = ax^2 + bx + c$:

Formula Used: $a > 0 \iff$ Parabola opens upwards; $a < 0 \iff$ Parabola opens downwards

Therefore an upward-opening parabola requires $\boxed{a > 0}$. [1M]

Q3. The zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ are: [1]

(A) $\frac{3}{2}, -\frac{1}{3}$ (B) $-\frac{3}{2}, \frac{1}{3}$ (C) $\frac{3}{2}, \frac{1}{3}$ (D) $-\frac{3}{2}, -\frac{1}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Rewriting in standard form $6x^2 - 7x - 3$, we know the middle-term splitting identity:

Formula Used: $6x^2 - 7x - 3 = (2x - 3)(3x + 1)$

Setting to 0 gives $2x - 3 = 0 \implies x = \frac{3}{2}$ or $3x + 1 = 0 \implies x = -\frac{1}{3}$, i.e., $\boxed{\frac{3}{2}, -\frac{1}{3}}$. [1M]

Q4. If zeroes of the quadratic polynomial $ax^2 + bx + c$ are equal, then: [1]

(A) c and a have opposite signs **(B)** c and b have opposite signs

(C) c and a have the same sign **(D)** c and b have the same sign

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

We know the equal-roots condition ($a \neq 0$, assuming non-zero $c \neq 0$):

Formula Used: $D = b^2 - 4ac = 0 \implies b^2 = 4ac$

Since $b^2 \geq 0$ and $a, c \neq 0$, we obtain $ac > 0$, meaning $\boxed{\text{(C) } c \text{ and } a \text{ have the same sign}}$. [1M]

Q5. Assertion-Reason: [1]

Assertion (A): If α, β are zeroes of $x^2 - 6x + k$ and $3\alpha + 2\beta = 20$, then $k = -16$.

Reason (R): For $x^2 - 6x + k$, $\alpha + \beta = 6$ and $\alpha\beta = k$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know Vieta's relations from Reason **(R)**:

Formula Used: $\alpha + \beta = 6, \qquad \alpha\beta = k$

Subtracting $2(\alpha + \beta) = 12$ from $3\alpha + 2\beta = 20$ yields $\alpha = 8$, hence $\beta = 6 - 8 = -2$. Then $k = \alpha\beta = (8)(-2) = -16$. Both **(A)** and **(R)** are true and **(R)** explains **(A)**. [1M]

SECTION B: Very Short Answer Questions — MATHS MANTHAN | 3 × 2 = 6 Marks

Q6. If the product of zeroes of $f(x) = x^2 - 4x + k$ is 3, find the value of k. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $f(x) = x^2 - 4x + k$ with $ax^2 + bx + c$, we have $a = 1$ and $c = k$. We know:

Formula Used: Product of zeroes $= \frac{c}{a}$

Setting $\frac{k}{1} = 3$ yields $\boxed{k = 3}$. **[2M]**

Q7. If α, β are zeroes of $f(x) = x^2 - p(x+1) - c$, prove that $(\alpha+1)(\beta+1) = 1 - c$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Expanding $f(x) = x^2 - px - (p+c)$, we have $a = 1, b = -p, c_0 = -(p+c)$. We know Vieta's formulas:

Formula Used: $\alpha + \beta = p, \qquad \alpha\beta = -(p+c)$

Expanding the LHS:

$$(\alpha+1)(\beta+1) = \alpha\beta + \alpha + \beta + 1 = -(p+c) + p + 1 = \boxed{1 - c = \text{RHS.}} \qquad \textbf{[2M]}$$

Q8. If one zero of $3x^2 + 8x + k$ is reciprocal of the other, find the value of k. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the zeroes be α and $\frac{1}{\alpha}$. We know Vieta's product formula:

Formula Used: $\alpha\beta = \frac{c}{a}$

Here $a = 3, c = k$, so $\alpha \cdot \frac{1}{\alpha} = \frac{k}{3} \implies 1 = \frac{k}{3} \implies \boxed{k = 3}$. **[2M]**

SECTION C: Short Answer Questions — MATHS MANTHAN | 3 × 3 = 9 Marks

Q9. If α and β are zeroes of $f(x) = 2x^2 + 11x + 5$, find the values of: **(i)** $\alpha^2 + \beta^2$ **(ii)** $\frac{1}{\alpha} + \frac{1}{\beta}$ **(iii)** $\alpha^3 + \beta^3$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $f(x) = 2x^2 + 11x + 5$, $\alpha + \beta = -\frac{11}{2}$ and $\alpha\beta = \frac{5}{2}$. We know:

Formula Used: $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta, \qquad \alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

(i) $\alpha^2 + \beta^2 = \left(-\frac{11}{2}\right)^2 - 2\left(\frac{5}{2}\right) = \frac{121}{4} - 5 = \frac{121 - 20}{4} = \boxed{\frac{101}{4}}$, **[1M]**

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-11/2}{5/2} = \boxed{-\frac{11}{5}}$, **[1M]**

(iii) $\alpha^3 + \beta^3 = \left(-\frac{11}{2}\right)^3 - 3\left(\frac{5}{2}\right)\left(-\frac{11}{2}\right) = -\frac{1331}{8} + \frac{165}{4} = \frac{-1331 + 330}{8} = \boxed{-\frac{1001}{8}}$. **[1M]**

Q10. If α, β are the zeroes of the quadratic polynomial $ax^2 + bx + c$, express $\dfrac{\alpha^2}{\beta} + \dfrac{\beta^2}{\alpha}$ in terms of a, b, c. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

For $ax^2 + bx + c$ $(a \neq 0, c \neq 0)$, we have $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$. We know:

Formula Used: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Simplifying over common denominator $\alpha\beta$:

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta} \qquad \text{[1.5M]}$$

$$= \frac{\left(-\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right)}{\frac{c}{a}} = \frac{\frac{-b^3+3abc}{a^3}}{\frac{c}{a}} = \boxed{\frac{3abc - b^3}{a^2 c}}. \qquad \text{[1.5M]}$$

Q11. If α, β are zeroes of $f(x) = 2x^2 - 5x + 7$, find a quadratic polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

From $f(x) = 2x^2 - 5x + 7$, $\alpha + \beta = \frac{5}{2}$ and $\alpha\beta = \frac{7}{2}$. We know:

Formula Used: $g(x) = k\left[x^2 - S'x + P'\right]$, $\quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

Computing S' and P' for $\alpha' = 2\alpha + 3\beta$ and $\beta' = 3\alpha + 2\beta$:

$$S' = (2\alpha + 3\beta) + (3\alpha + 2\beta) = 5(\alpha + \beta) = 5\left(\frac{5}{2}\right) = \frac{25}{2}, \qquad \text{[1M]}$$

$$P' = 6\alpha^2 + 13\alpha\beta + 6\beta^2 = 6(\alpha + \beta)^2 + \alpha\beta = 6\left(\frac{25}{4}\right) + \frac{7}{2} = \frac{75}{2} + \frac{7}{2} = 41. \qquad \text{[1M]}$$

Setting $k = 2$ yields $g(x) = 2\left(x^2 - \frac{25}{2}x + 41\right) = \boxed{2x^2 - 25x + 82}$. [1M]

SECTION D: Long Answer / HOTS Questions — MATHS MANTHAN | 2 × 5 = 10 Marks

Q12. If α, β are the zeroes of $f(x) = x^2 - 4x + 1$, evaluate: **(i)** $\alpha^2 + \beta^2$ **(ii)** $\alpha^3 + \beta^3$ **(iii)** $\alpha^4 + \beta^4$ **(iv)** $\dfrac{\alpha}{\beta^2} + \dfrac{\beta}{\alpha^2}$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

From $f(x) = x^2 - 4x + 1$, we have $\alpha + \beta = 4$ and $\alpha\beta = 1$. [1M]

We know the symmetric power identities:

Formula Used: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, $\quad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

(i) $\alpha^2 + \beta^2 = (4)^2 - 2(1) = 16 - 2 = \boxed{14}$, [1M]

(ii) $\alpha^3 + \beta^3 = (4)^3 - 3(1)(4) = 64 - 12 = \boxed{52}$, [1M]

(iii) $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = (14)^2 - 2(1)^2 = 196 - 2 = \boxed{194}$, [1M]

(iv) $\dfrac{\alpha}{\beta^2} + \dfrac{\beta}{\alpha^2} = \dfrac{\alpha^3 + \beta^3}{(\alpha\beta)^2} = \dfrac{52}{1^2} = \boxed{52}$. [1M]

Q13. If the quadratic polynomial $f(x) = ax^2 + bx + c$ satisfies $f(1) = 0$, $f(2) = 3$, and $f(3) = 8$, determine the coefficients a, b, c. Hence find the zeroes of the polynomial. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Substituting $x = 1, 2, 3$ into $f(x) = ax^2 + bx + c$:

$$f(1) = 0 \implies a + b + c = 0 \quad \text{— (1)}$$
$$f(2) = 3 \implies 4a + 2b + c = 3 \quad \text{— (2)}$$
$$f(3) = 8 \implies 9a + 3b + c = 8 \quad \text{— (3)} \qquad \textbf{[1M]}$$

We know the elimination method and difference-of-squares factorisation:

Formula Used: $x^2 - 1 = (x - 1)(x + 1)$

Subtracting (1) from (2) and (2) from (3) to eliminate c:

$$3a + b = 3 \quad \text{— (4)}, \qquad 5a + b = 5 \quad \text{— (5)}. \qquad \textbf{[1M]}$$

Subtracting (4) from (5): $2a = 2 \implies \boxed{a = 1}$. Substituting $a = 1$ into (4): $3(1) + b = 3 \implies \boxed{b = 0}$.
Substituting into (1): $1 + 0 + c = 0 \implies \boxed{c = -1}$. **[1.5M]**
Zeroes: $f(x) = x^2 - 1 = (x - 1)(x + 1) = 0 \implies$ Zeroes: $\boxed{1 \text{ and } -1}$. **[1.5M]**

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